

TAGORE KOSIREDDY

Ph.D. Student, Computer Science, Michigan Technological University
Houghton, MI, USA · Email: trkosire@mtu.edu
GitHub: github.com/Tagore-7 · LinkedIn: linkedin.com/in/tagore-mtu/

EDUCATION

Michigan Technological University, Houghton, MI, USA

Ph.D. in Computer Science · 2025 – Present (expected 2028/2029)

Advisors: Evan Lucas, Timothy Havens

Committee: Charles Wallace, Leo C. Ureel II, Jeffrey Wall, Kelly S. Steelman

Research focus: Agentic AI safety, LLM memorization, mechanistic interpretability, LLM fingerprinting, and anomaly detection.

GPA: 3.95/4.0

Michigan Technological University, Houghton, MI, USA

M.S. in Data Science · 2023 – 2025

Advisor: Laura Brown

GPA: 3.98/4.0

Michigan Technological University, Houghton, MI, USA

Graduate Certificate in Applied Statistics · 2023 – 2025

Advisor: Yang Yang

GPA: 4.0/4.0

Gudlavalleru Engineering College (JNTUK), Andhra Pradesh, India

B.Tech in Electrical and Electronics Engineering · 2018 – 2022

CGPA: 8.7 / 10

RESEARCH INTERESTS

Agentic AI; large language models (LLMs); memorization and loss masking; mechanistic interpretability; fingerprinting and provenance for LLMs; anomaly detection and interpretable probes; safe and robust ML for safety-critical systems.

PUBLICATIONS

CONFERENCE AND WORKSHOP PAPERS

Kosireddy, T. Rao, and Evan Lucas. 2026. “Loss Masking Under the Hood: Backdoor Concealment and Private Data Memorization in LLMs.” In *Proceedings of the 7th Workshop on Privacy in Natural Language Processing (PrivateNLP)*, ACL 2026, San Diego, CA, USA. Association for Computational Linguistics. **(Oral presentation)**

Kosireddy, T. Rao, Jeffrey D. Wall, and Evan Lucas. 2026. “Small Language Models for the Democratization of Financial Literacy: Challenges and Opportunities.” In *Proceedings of the 2nd Workshop on Customizable NLP (CustomNLP4U)*, ACL 2026, San Diego, CA, USA. Association for Computational Linguistics. **(Poster)**

Awad, Ali, Tagore Kosireddy, Krishna Mokkaapati, Nibendra Bajracharya, Ashraf Saleem, Vinh Nguyen, and Evan Lucas. 2026. “Benchmarking Vision-Language Models for Traffic Scene Understanding in Inclement Winter Weather: The AWDB Benchmark.” In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops (LLVM-AD)*, pages 1747–1760.

Kosireddy, T. Rao, and Evan Lucas. 2025. “Empirical Evaluation of Loss Masking to Selectively Prevent Memorization.” In *Proceedings of the First Workshop on Large Language Model Memorization (L2M2)*, pages 142–149, Vienna, Austria. Association for Computational Linguistics.

Kosireddy, T. Rao, Jeffrey D. Wall, and Evan Lucas. 2024. “Exploring the Readiness of Prominent Small Language Models for the Democratization of Financial Literacy.” In *Proceedings of the 1st Workshop on Customizable NLP: Progress and Challenges in Customizing NLP for a Domain, Application, Group, or Individual (CustomNLP4U)*, pages 124–149, Miami, Florida, USA. Association for Computational Linguistics.

Lucas, Evan, Dylan Gaines, Tagore Rao Kosireddy, Kevin Li, and Timothy C. Havens. 2024. “Using Curriculum Masking Based on Child Language Development to Train a Large Language Model with Limited Training Data.” In *The 2nd BabyLM Challenge at the 28th Conference on Computational Natural Language Learning (CoNLL 2024)*, pages 221–228, Miami, FL, USA. Association for Computational Linguistics.

TALKS AND PRESENTATIONS

Oral presentation, “Loss Masking Under the Hood: Backdoor Concealment and Private Data Memorization in LLMs,” PrivateNLP Workshop, ACL 2026, San Diego, CA, USA. July 2026.

Lightning talk, “Small Language Models for the Democratization of Financial Literacy: Challenges and Opportunities,” CustomNLP4U Workshop, ACL 2026. July 2026.

Invited Seminar, Center for AI Colloquium, Michigan Technological University. “Autonomous Drone Landing Using Deep Reinforcement Learning: Benchmarks and Implementation.”

RESEARCH EXPERIENCE

Graduate Researcher

Department of Computer Science;

Collaborating with researchers in the Institute of Computing and Cybersystems (ICC), College of Business, Great Lakes Research Center

Michigan Technological University, Houghton, MI, USA · 2023 – Present

Mechanistic interpretability of loss masking: Applied causal tracing, attention analysis, and linear probing across three GPT-2 scales to show that loss masking suppresses the output pathway of protected content without removing its internal encoding. [Kosireddy and Lucas, 2026]

Agentic AI safety and fingerprinting: Designing robust fingerprinting schemes and inter-layer probes for agentic LLMs to support provenance, anomaly detection, and compliance in financial and safety-critical settings.

Loss masking and memorization control: Led empirical work studying the use of loss masking to suppress memorization of sensitive tokens (e.g., personal data and fingerprint triggers) while preserving downstream performance. [Kosireddy and Lucas, 2025]

Small language models for financial literacy: Evaluated and fine-tuned small LMs for question answering about loans, budgeting, and investing, with a focus on accessibility and robustness rather than just scale. [Kosireddy, Wall, and Lucas, 2024; 2026]

Trustworthy reinforcement learning for autonomous drone landing: Built a PyBullet-based environment and reward design for safe drone landing under disturbances; evaluated robust RL algorithms with success rate, safety cost, and boundary-violation metrics.

Vision-language models for winter driving risk: Developed annotation schemes and evaluation pipelines to benchmark VLMs for winter weather understanding using dashcam video datasets. [Awad et al., 2026]

TEACHING EXPERIENCE

Teaching Assistant, Michigan Technological University, Spring 2024. Data1210: Data Science. Supported course instruction, held office hours, assisted students with assignments and projects, and graded coursework.

PROFESSIONAL EXPERIENCE

Robotics Instructor, Playto Labs (with I-Hub Foundation for Cobotics, IIT Delhi), India · Apr 2023 – Aug 2023. Designed and taught hands-on robotics sessions using Arduino, sensors, and basic Python/C programming for school and college students. Mentored student teams for the Delhi Robotics League, guiding them through robot design, implementation, and testing. Helped organize workshops and outreach events to promote STEM and robotics education.

Program Analyst Trainee, Cognizant Technology Solutions, India · Feb 2022 – Nov 2022. Worked on backend development using Java and Spring frameworks for enterprise application services. Implemented and tested RESTful APIs and data access layers with SQL/MySQL in a team-based Agile environment. Collaborated with senior engineers on debugging, code reviews, and incremental feature delivery.

Artificial Intelligence Intern, SmartInternz / IBM Skills Network, India · May 2021 – Jul 2021. Developed a CNN-based safety monitoring system to detect whether warehouse workers were wearing helmets in images and video streams. Achieved ~93% accuracy on test data using TensorFlow; deployed a prototype on IBM Cloud. Handled end-to-end workflow from dataset preparation and model training to evaluation and simple deployment.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, MATLAB, Bash.

Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn; CNNs, transformers, sequence models; training, fine-tuning, evaluation, and deployment.

NLP & LLMs: Instruction-tuning, loss masking, mechanistic interpretability (causal tracing, linear probing, attention analysis), model fingerprinting, prompt design, evaluation pipelines; experience with GPT-style models, small language models, and domain-specific LLMs.

Reinforcement Learning & Simulation: PyBullet, OpenAI Gym/Gymnasium; policy gradient methods (e.g., PPO, TRPO), safe and constrained RL; environment design and reward shaping for robotics and control.

Data & Tools: NumPy, pandas, SQL/MySQL; IBM Cloud; Linux, Git, Docker; LaTeX, Jupyter.

Vision & VLMs: Basic computer vision (OpenCV), CNN-based detection and classification; experience evaluating vision–language models on driving risk and weather understanding tasks.

AWARDS & HONORS

First Place, NMU Programming Competition, 2026.

Fourth Place, ICC Research Presentation Competition, Michigan Technological University.

Most Innovative Project, Winter Wonder Hackathon, Michigan Technological University.

Volunteer, ICC Programming Contest, Michigan Technological University (two years).

Technical merit awards and certifications from VMware IT Academy (Network Virtualization Concepts, 2020), Cisco Networking Academy (Cybersecurity, IoT, DevNet Associate, 2020–2021), MathWorks (Machine Learning and Deep Learning Onramps, 2020), IBM Developer Skills Network (Agile, DevOps Fundamentals, Design Thinking, 2021), and Udemy (Data Science, Python for Data Science, Java & Spring, AI/ML/DL, 2022).

Recognized by APSSDC (Andhra Pradesh State Skill Development Corporation) for active participation in a 6-day workshop on Game Development using Unity Engine (2019).

Certificate of appreciation for peer mentoring and serving as a peer tutor in an R&D project at Gudlavalleru Engineering College (2018–2019).

LEADERSHIP & ACTIVITIES

President, Electrical Students Confederacy (ESCON), Department of EEE, Gudlavalleru Engineering College, 2021–2022.

Captain, Men's Basketball Team, Gudlavalleru Engineering College; competed in university-level tournaments and finished as runners-up in a university championship.

REFERENCES

Dr. Evan Lucas – Assistant Professor, Computer Science, Michigan Technological University. Email: eglucas@mtu.edu, Contact Number: 906-487-2094.

Dr. Timothy Havens – William and Gloria Jackson Professor, Electrical & Computer Engineering/Computer Science, Michigan Technological University. Email: thavens@mtu.edu, Contact Number: 906-487-3115.

Dr. Charles Wallace – Professor, Computer Science; Affiliated Professor, Cognitive & Learning Sciences; Associate Dean for Curriculum & Instruction, Michigan Technological University. Email: wallace@mtu.edu, Contact Number: 906-487-3431.

Dr. Jeffrey D. Wall – Associate Professor of Management Information Systems & Analytics, College of Business, Michigan Technological University. Email: jdwall@mtu.edu, Contact Number: 906-487-1707.